Sindhudurg Shikshan Prasarak Mandal's College of Engineering, Kankavli

A/P-HarkulBudruk,Tal- Kankavli. Dist-Sindhudurg.Maharashtra.416602
PHONE: 91-2367-299214(office)/ 299112
Departmentsof
Computer Engineering, Computer Science & Engineering, Computer Science & Engineering (AIML)

Experiment No -1

Year: 2026-27	Class/Div.-TE	SEM: V
Course Code: 2115311	Course Name: Open Source Technology	
Branch:ComputerEngineering,Computer Science&Engineering&AIML(ComputerScienceEngineering)		

Aim	Design a professional technical document in LibreOffice Writer; Build an automated calculation sheet using functions in LibreOffice Calc; Perform Layer masking, colour adjustments, and asset export using GIMP.
Software Requirements	Libre Office Writer, Libre Office Calc, Libre Office Impress, GIMP
Theory	LibreOffice Writer **1.Structure&Layout** •**Headingsv s. SubPoints:**Headings use large, prominent typography (18-20pt, bold, underlined) and localized color fills to establish a clear reading hierarchy. Sub points scale down (14pt) and use bullets (Shift + F12) or numbers to organize raw notes into digestible arrays. • **Text Formatting:** Standard body text is kept at a readable 11-12pt size, strictly black for formal documents, and often **Justified** (Ctrl + J) to align text perfectly to both margins. You can use Superscript (Ctrl + Shift + P) or Subscript (Ctrl + Shift + B) for scientific notations or formulas. **2. Media, Tables & Shapes** • **Image Integration:** Beyond copy-paste, you can right-click images to **Crop** out dead space or change **Wrap** settings (like *Page Wrap* or *Optimal Wrap*), which forces text to flow smoothly around the image instead of colliding with it. • **Table Formatting (Ctrl + F12):** Tables organize data grids. You can customize header cell text (bold, color), apply distinct background fills to rows, add bullets inside single cells, and strip away default borders for a clean, borderless layout. **Shape Formatting:** Accessed via View > Toolbars > Drawing. You can draw geometric vectors, apply multi-tone color gradients, and double-click them to type text directly inside the shape boundaries. • **Page Setup:** Handled globally under Format > Page Style... to configure margins, page orientation (Portrait/Landscape), paper size, or

Subject Incharge- M.S.Jamsandekar.

split text into newspaper-style Columns. Use Format > Watermark... to add transparent safety text (like "DRAFT") behind your work.

3. Quick Global Editing Shortcuts

- Ctrl + F (Find): Instantly scan the entire document to highlight a specific keyword.
- Ctrl + H (Replace): Globally swap out every instance of a specific typo, term, or variable with a single click.
- Ctrl + A (Select All): Grabs every text and image asset in the document at once to apply global font or style changes.

LibreOffice Calc

1. Grid Mechanics & Data Entry

- Spr asheet are made of columns (letters) and

  TheAddressSystem:
  rows (numbers). The intersection creates a specific cell address (e.g., A1 is top-left). Calc automatically left-aligns text data and right-aligns numeric data.
- **AutoFill (The Magic Square):** When you click a cell, a tiny black square appears in the bottom-right corner. Dragging this handle down automatically continues sequential patterns like dates, days of the week, or serial numbers.
- **Cell Formatting (Ctrl + 1):** Allows you to convert raw numbers into readable data types like Currency ($), Percentages (%), or formatted Dates without altering the actual underlying value. Double-clicking line dividers between column letters auto-fits the column width to your longest entry.

2. Formulas & Functions

TheGolden Rule: Every mathematical instruction *must* start with an equals sign (=). If you omit it, Calc treats the entry as plain text.

- **Manual Math:**Uses standard operational symbols between specific cells:
  - Addition: =A1+B1
  - Subtraction: =A1-B1
  - Multiplication: =A1*B1
  - Division: =A1/B1

- **Built-in Functions:** Used to calculate massive cell ranges cleanly without typing manual mathematical operators:
  - =SUM(A1:A10) — Total math addition of all values inside the range.
  - =AVERAGE(A1:A10) — Calculates the exact mathematical mean.
  - =COUNT(A1:A10) — Returns a count of how many cells in that range contain actual numerical values.

3. Organization& Visualization

- **Sorting &Filtering:** Use Data > Sort to instantly reorder entire worksheets alphabetically or numerically. Use Data > AutoFilter to add dropdown arrows to your header row, allowing you to filter out specific records (e.g., hiding all data except "Groceries").
- **Charts:** Selecting a data range and clicking Insert > Chart opens a tool to generate visual representations. Master **Column/Bar charts** to compare independent items, and **Pie charts** to display parts of a whole (like a budget share).

LibreOffice Impress

1. Interface Navigation

- **SlidesPane(LeftSide):** A vertical thumbnail list of your entire deck. Usethis area toquickly duplicate, delete, rename, or drag slides to reorder your narrative.
- **Workspace (Center):** The main design canvas where you position your elements.
- **Sidebar (Right Side):** The primary control dashboard for assigning text properties, selecting preset slide layouts, adjusting animations, and applying slide transitions.

2. Slide Mechanics & Content

- **UsingLayouts:** Instead of drawing text fields manually, use the Sidebar's layout panel to pick pre-set arrangements optimized for titles, bullets, or side-by-side comparisons.
- **TheMasterSlide(View>MasterSlide):**Theabsoluteblueprint of your presentation. Modifying the background color, font choice, or adding a company logo here instantly forces that change across every single slide in the deck, saving hours of manual editing.

- **Content Restraint:** Slides should handle minimal text load (aim for a maximum rule of 5 to 8 lines per slide) so the audience reads the content cleanly. Embed data charts, shapes, and images (Insert > Image) to replace text blocks with visuals.

3. Display Modes & Motion

- **The Four Views:**
  - *Normal:*Standardviewforediting slide layouts.
  - *Outline:*Drops allvisualgraphics to show only the text, perfect for rapidbrainstormingandtyping out scripts.
  - *Slide Sorter:* Displaysyourdeck as a broad grid of cards, makingiteasy toseethepacing and move large slide blocks around.
  - *Notes View:* A splitscreentotype private speaker notes. These remaininvisible toyouraudience and only show up on your presenter monitor.
  - **O**

- **Transitions vs. Animations:**
  - *Slide Transitions*arethevisual animations that occur **between**slides asyouprogress (e.g., Fade, Push, Wipe).
  - *Animations* dictatehowspecific independent objects (like a single bullet pointorimage)behave **inside** a single slide workspace.

Digital Imaging Fundamentals in GIMP

1. Navigating the Workspace

- **The Canvas:** Thecentralzonewhere your graphic image is actively rendered. Contains your active operational brushes, selectionlassoes, croputilities,and spatial transformation tools.

- **TheTLaoyoelbrsoxD(oLcekft(RSiigdhet):Side):**Yourlayout overview: evera ccidentally closes,restoreit instantly via Windows > Dockable Dialogs > Layers.

2. Understanding Layers

	• **The Layer Stack:** Think of Layers like transparent acetate sheets stacked perfectly on top of one another. Instead of destructive painting on one single flat canvas, you isolate your background, text, and main subject elements onto their own independent Layers. • **Design Control:** This structure means you can move, resize, distort, or erase oneLayer withouttouching or accidentally ruining any of the pixel dataontheother sheets.Toggle a Layer's visibility on or off by clicking the **Eye** iconnext to it inthedock. **3. Mastering Non-Destructive Editing** • **Destructivevs.Non-Destructive:** "Destructive" editing permanently overwrites original image pixels (like using the eraser tool or cropping a base photo). Non-destructive editing preserves your raw image data completely so you can revert mistakes later. • **Duplicating Layers:** Before applying any heavy filter, color grading, or edit to a photo, right-click the base Layer and select **Duplicate Layer**. Keep the original at the bottom of your stack, locked and untouched as a master backup. • **LayerMasks(TheProAlternativetoErasing):**Insteadofusingthe destructive Eraser tool to remove backgrounds, add a Layer Mask to your active Layer. o Painting with **Black** ink on the mask hides those pixels (making them transparent). o Painting with **White** ink on the mask reveals them again. If you over-erase an edge by accident, you don't have to hit undo—you simply paint over the mistake with a white brush to instantly restore the original pixels.
Procedure	Image 1: Resume Layout (LibreOffice Writer) 1. Page Setup & Banners: o Open LibreOffice Writer. o Use the Drawing Tools (Insert > Shape > Basic Shapes > Right Triangle) to draw blue diagonal accents at the top-left and bottom-right corners. Set their fill color to blue. 2. Text & Profile Image: o Center-align and bold the title RESUME. o Type out the PERSONAL DETAILS section.

- Go to Insert > Image to place the profile photo at the top right

3. Tables & Bullet Points:
   - Go to Table > Insert Table (3 rows × 3 columns) to populate the EDUCATIONAL QUALIFICATION table.
   - Enter the text for Technical Skills, Industrial Training, and Certifications, applying Bullet Lists (Shift + F12) where needed.
4. Declaration:
   - Type out the declaration text at the bottom. Save or export as PDF.

Image 2: Result Sheet, Calendar & Timetable (LibreOffice Calc)

1. Sheet Setup: Open LibreOffice Calc.
2. Table 1 (Student Result Sheet):
   - Merge top cells for the title banner and fill with dark blue background and white text.
   - Add student records and apply conditional formatting/cell fill colors (e.g., green for high marks/PASS, red for low marks/FAIL).
   - Use formulas like =SUM() for Total and =AVERAGE() or simple division for Percentage.
3. Table 2 (February Calendar & Collage):
   - Insert the group image on the left side (Insert > Image).
   - Create a 7×6 cell grid on the right for the February calendar days, setting border lines.
   - Add a text note area below the calendar.
4. Table 3 (Class Timetable):
   - Create a timetable matrix with custom cell background colors (yellow header bar).
   - Merge vertical cells across time slots for LUNCH BREAK and set text orientation/alignment to vertical.

Image 3: Presentation Portfolio (LibreOffice Impress)

1. Slide Setup: Open LibreOffice Impress and select a standard 16:9 blank template.
2. Slide 1 (Header Slide):
   - Insert a blue horizontal banner shape for the "PORTFOLIO" title block.
   - Crop the profile picture into a circle/ellipse and place it next to personal identification details.
3. Slide 2 (About Me & Education):
   - Insert a top header bar ("About Me & Education").
   - Add bullet points for "About Me" and "Skills".
   - Insert a 4×3 table in the middle for qualifications.

4. Slide 3 (Projects & Contact):
    - Add the header banner ("Projects & Contact").
    - Divide the slide using a vertical line shape (Insert > Shape > Line).
    - Add bullet lists on the left (Projects) and right (Hobbies & Strengths).

Image 4: Image Editing / Portfolio Poster (GIMP)

1. Canvas Setup:
    - Open GIMP and create a new project (File > New) set to A4 or standard screen dimensions with a light grey background (#E6E6E6).
2. Creating Header Banners:
    - Use the Rectangle Select Tool (R) to select header strip regions.
    - Use the Bucket Fill Tool (Shift + B) to fill the selection with dark blue (#1A4B75).
3. Photo Masking & Text Placement:
    - Import the profile photo as a new layer (File > Open as Layers).
    - Apply an Ellipse Select circular mask on the photo layer to create a round profile frame.
    - Use the Text Tool (T) to add titles, section headings, and body copy using standard sans-serif fonts.
4. Layout Lines & Export:
    - Draw vertical/horizontal divider lines using the Pencil/Paintbrush Tool holding Shift for straight lines, or using guides.
    - Go to File > Export As to save the final composite image as JPG/PNG.

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Output

1. LibreOffice Writer



Shruti Jitendra Kasle

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Skills

- **Programming Languages**: Python, Java, Kotlin
- **Database:** MySQL, SQLite
- **Web Technologies**: HTML, CSS
- **Tools & IDEs:** Android Studio, VS Code
- **Other Skills:** OOPs, Data Structure(basics)

Languages

- English
- Hindi
- Marathi

Extra Curricular

- Participated in college technical events and workshops.
- Participated in coding competitions / hackathons.
- Volunteered in college events.

Profile

B.E. CSE (AI & ML) student with a strong foundation in Python, Machine Learning, Data Structures, and Web Development. Passionate about building practical technology solutions and continuously developing technical skills through projects and hands-on learning.

Education

Sindhudurg Shikshan Prasarak Mandal's College Of Engineering, Kankavli
T.E., CGPA: 8.8 (Till Sem 4)
Sindhudurg, Maharashtra — Aug 2026

Projects

SAARTHI Navigator(Colllege Info App)

- Android applocation which provides information about colleges in the Konkan region.
- **Tech Stack :** Kotlin, XML, Firebase(Basic)
- **Features :** College listing, search, details, contact info.

StudentDB (Database Project)

- Designed a database system for student and course management.
- **Tech Stack :** SOLite, SQL.
- Implemented tables, queries, and relationships.

Certifications

1. *Python Programming Certification*
2. *Java Certification*
3. *AWS Certified Solutions Architect – Associate*

Personal Information

- **Date of Birth:** 4th Jan 2007
- **Gender:** Female
- **Nationality:** Indian

Accomplishments

- Participated in college technical events and workshops.
- Participated in coding competitions / hackathons.
- Volunteered in college events.

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2. LibreOffice Calc

Sr. No.	Roll No.	Student Name	Operating Syster	Database Management Syste	Computer Network	AIML	Open Elective	TOTAL (500	PERCENTAGE (	GRADE	RESULT
1	CE001	Yogita Mestry	89	92	87	94	1	CE001	Yogita Mestry	93	94
2	CE002	Siddhesh Patil	78	85	80	88	2	CE002	Siddhesh Patil	89	91.5
3	CE003	Sanika Parab	68	74	71	79	3	CE003	Sanika Parab	80.5	83.5
4	CE004	Omkar Sawant	55	64	60	71	4	CE004	Omkar Sawant	73.5	77.9
5	CE005	Pranav Kadam	45	58	50	62	5	CE005	Pranav Kadam	64.5	68.8
6	CE006	Rutija Jadhav	39	45	42	48	6	CE006	Rutija Jadhav	49.5	51.9
7	CE007	Aditya More	34	40	38	45	7	CE007	Aditya More	47	50.1
8	CE008	Sakshi Naik	92	93	95	96	8	CE008	Sakshi Naik	97.5	98.9
9	CE009	Atharva Gurav	81	83	79	86	9	CE009	Atharva Gurav	85	86.1
10	CE010	Mayuri Shinde	63	69	66	72	10	CE010	Mayuri Shinde	73.5	75.9



February 2026

MON	TUE	WED	THU	FRI	SAT	SUN
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28

Time Table

DAY	Class	09:30-10:30	10:30-11:30	11:30-11:45	11:45-12:45	12:45-1:45	01:45-02:30	02:30-03:30	03:30-04:40
Mon	TE AIML	CN/MDM/OE		Short Break	MDM/OE/CN		Lunch Break	CN	IKS
Tue		OE/CN/MDM			SMLDS	MDM		SMLDS/AISC/ASDD	
Wed		SMLDS	CN		AISC/ASDD/SMLDS			MDM	AISC
Thu		AISC	ASDD		IKS	MDM		ASDD/SMLDS/AISC	
Fri		ASDD	SMLDS		CN	AISC		ASDD	IKS
Sat		ALSC	SMLDS		AISC	NPTEL Activity		NPTEL Activity	

Subject Incharge- M.S.Jamsandekar.

3. LibreOffice Impress



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4. GIMP



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Conclusion	This experiment successfully demonstrates the power of integrating open-source tools (LibreOffice Writer, LibreOffice Calc, LibreOffice Impress, GIMP) to execute a professional, end-to-end technical workflow.

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